29(2) $u=f(\frac{y}{x})=f(t)$

$\therefore u_x=f'(t)(-\frac{y}{x^2})$

$u_{xx}=f''(t)\cdot\frac{y^2}{x^4}+f'(t)\cdot(2\frac{y}{x^3})$

$u_y=f'(t)\cdot\frac{1}{x}$

$u_{yy}=f''(t)\frac{1}{x^2}$

$\because u_{xx}+u_{yy}=0 \quad \therefore f''(t)\cdot\frac{x^2+y^2}{x^4}+f'(t)\cdot 2\frac{y}{x^3}=0$

$\therefore f''(t)\cdot(1+t^2)+f'(t)\cdot 2t=0$

$\therefore [f'(t)(1+t^2)]'=0$

$\therefore f'(t)=\frac{C_1}{1+t^2} \quad \therefore f(t)=C_1\arctan t+C_2$

$\therefore u=C_1\arctan t+C_2$ (C_1,C_2为任意常数)

30.(2) 解: $f'(x)=\frac{x^2+y^2-2y^2}{(x^2+y^2)^2}+i\frac{-2xy}{(x^2+y^2)^2}$

$\therefore f(z)=\int(\frac{x^2-y^2}{(x^2+y^2)^2}+i\frac{-2xy}{(x^2+y^2)^2})dz+C$

$=\int\frac{\bar{z}^2}{z^2\cdot\bar{z}^2}dz+C=\int\frac{1}{z^2}dz+C=-\frac{1}{z}+C$

$\because f(1)=0 \quad \therefore f(z)=-\frac{1}{z}+\frac{1}{z}$

31 $\because v_{xx}+v_{yy}=0$

$\therefore p^2e^{px}\sin y-e^{px}\sin y=0 \quad \therefore p=\pm 1$

$p=1$时 $f(z)=\int(e^x\cos y+ie^x\sin y)dz+C_1$

$=e^z+C_1$

$p=-1$时 $f(z)=\int(e^{-x}\cos y-ie^{-x}\sin y)dz+C_2$

$=-e^{-z}+C_2$

(C_1,C_2)均为任意常数

练习1: $f(z)$解析 $\Rightarrow u'(xy)\cdot y=v_y \quad u'(xy)\cdot x=-v_x$

$\therefore u''(xy)\cdot y^2=v_{xy} \quad u''(xy)\cdot x^2=-v_{xy}$

$\therefore u''(xy)\cdot(x^2+y^2)=0 \quad \therefore u''(xy)=0$

$\therefore$ 设 $u=C_1xy+C_2$ (C_1,C_2均为实常数)

$\therefore v_y=C_1y \quad v_x=-C_1x$

$\therefore v=\frac{C_1}{2}y^2+g(x)$

$\therefore v_x=g'(x)=-C_1x \quad \therefore g(x)=-\frac{C_1}{2}x^2+C_3$

$\therefore v=\frac{C_1}{2}y^2-\frac{C_1}{2}x^2+C_3$ (C_3为实常数)

$\therefore f=C_1xy+C_2+i\frac{C_1}{2}(y^2-x^2)+iC_3$

$=-i\frac{C_1}{2}(x^2-y^2+2ixy)+C_2+iC_3$

$=iAz^2+B$

$A\in\mathbb{R} \quad B\in\mathbb{C}$ 均为常数

练习2: (1) 当$n\to\infty$时

$\frac{\sin\frac{1}{n}}{n}\sim\frac{1}{n^2}$

$\because\sum_{n=1}^{\infty}\frac{1}{n^2}$绝对收敛

$\therefore\sum_{n=1}^{\infty}\frac{\sin\frac{1}{n}}{n}$绝对收敛

(2) $\because\frac{(\cos in)}{2^n}=\frac{e^n+e^{-n}}{2^{n+1}}=\frac{e^n}{2^{n+1}}+\frac{e^{-n}}{2^{n+1}}$

$\therefore\lim_{n\to\infty}|\frac{\frac{e^{n+1}}{2^{n+2}}}{\frac{e^n}{2^{n+1}}}|=\frac{e}{2}>1$

$\lim_{n\to\infty}|\frac{\frac{e^{-n-1}}{2^{n+2}}}{\frac{e^{-n}}{2^{n+1}}}|=\frac{1}{2e}<1$

$\therefore\frac{e^n}{2^{n+1}}$发散, $\frac{e^{-n}}{2^{n+1}}$绝对收敛

$\therefore\frac{\cos in}{2^n}$发散

(3) $\frac{e^{2ni}}{n^2}=\frac{\cos 2n+i\sin 2n}{n^2}$

$\because\frac{1}{n^2}$单调递减且趋于0

$\sum_{k=1}^{n}|e^{2ki}|=|\frac{e^{2i}(1-e^{2ni})}{1-e^{2i}}|$

$=|\frac{(\cos 2+\sin 2i)-(\cos(2n+2)+\sin(2n+2)i)}{1-\cos 2-\sin 2i}|$